

Basic Concept of Artificial Intelligence (AI)

What is Intelligence Composed of?

1. **Intelligence is composed of which of the following factors?**

- A) Perception, reasoning, memory
- B) Speed, strength, memory
- C) Algorithms, calculations, processing
- D) Repetition, execution, precision

Answer: A

2. **Which of the following is a component of human intelligence?**

- A) Ability to interact socially
- B) Physical strength
- C) A desire for rewards
- D) Only computational tasks

Answer: A

Learning What's Involved in AI

3. **Which of the following is NOT a typical step in learning AI?**

- A) Data collection
- B) Training models
- C) Programming specific task solutions
- D) Daily task delegation

Answer: D

4. **In AI, what does the term 'training' refer to?**

- A) Creating new data.
- B) Optimizing algorithms using data.
- C) Performing human-like tasks.
- D) Producing physical robots.

Answer: B

5. **Which of the following is used to evaluate AI models?**

- A) Cost-benefit analysis
- B) Performance metrics
- C) Hardware efficiency
- D) Power consumption rates

Answer: B

Application of AI

6. **AI applications are commonly used in which of the following industries?**

- A) Healthcare
- B) Entertainment
- C) Finance
- D) All of the above

Answer: D

7. **AI technology is commonly used in which of the following tasks?**

- A) Image recognition
- B) Autonomous driving
- C) Natural language processing
- D) All of the above

Answer: D

8. **Which of the following is an AI application in healthcare?**

- A) Medical diagnosis assistance
- B) Automated data entry
- C) Patient appointment scheduling
- D) Sending reminders

Answer: A

9. **What does cognitive modeling in AI aim to simulate?**

- A) Machine learning techniques
- B) Human thought and decision-making processes
- C) Mathematical calculations only
- D) Statistical analysis models

Answer: B

10. Which of the following is true about cognitive models?

- A) They help AI systems mimic human reasoning and actions.
- B) They are irrelevant to human-like intelligence.
- C) Cognitive models do not use data.
- D) Cognitive models only work for visual processing.

Answer: A

11. In cognitive modeling, what is a typical goal?

- A) Simulating a machine's internal wiring.
- B) Predicting human behaviors based on input data.
- C) Programming robots to perform specific actions.
- D) Solving complex mathematical theorems.

Answer: B

12. Which Python library is commonly used for machine learning tasks?

- A) TensorFlow
- B) Flask
- C) Django
- D) Kivy

Answer: A

13. Python is considered ideal for AI because:

- A) It is exclusively for scientific computing.
- B) It has good integration with other programming languages.
- C) It is the only language that supports AI.
- D) It is a compiled language.

Answer: B

Features of Python

14. Which of the following is NOT a feature of Python?

- A) Object-oriented programming support
- B) Large standard library
- C) Strong typing system
- D) Dynamic typing system

Answer: C

15. Which of the following makes Python particularly good for AI development?

- A) It is a compiled language.
- B) It has a low-level memory management system.
- C) It supports large community-driven libraries for AI.
- D) It requires manual memory management.

Answer: C

16. Which of the following Python libraries is particularly useful for data analysis in AI?

- A) Pandas
- B) Flask
- C) Pygame
- D) Keras

Answer: A

17. Python is considered a high-level language because it:

- A) Is close to hardware.
- B) Uses simple syntax that's easy to understand and write.
- C) Requires explicit memory management.
- D) Needs to be compiled into machine code.

Answer: B

18. How can you exit the Python interactive shell?

- A) By pressing Ctrl+Z or Ctrl+D
- B) By typing exit()
- C) By typing quit()
- D) All of the above

Answer: D

19. Which of the following is a necessary step to execute a Python script from the command line?

- A) Ensure the Python script has the .txt extension.
- B) Open the script with a text editor.
- C) Use the python command followed by the script's filename.
- D) Change the file format to .exe.

Answer: C

20. Why is an Integrated Development Environment (IDE) preferred for Python development?

- A) It integrates the code editor, debugger, and other useful tools.
- B) It automatically writes the code for you.
- C) It executes code without needing any Python installation.
- D) It is only used for advanced Python users.

Answer: A

21. Which feature of an IDE is particularly useful for AI development?

- A) Syntax highlighting and autocompletion
- B) Data visualization and debugging support
- C) Integrated terminal for script execution
- D) All of the above

Answer: D

22. In NLP, what does the process of Tokenization involve?

- A) Breaking down text into words, phrases, or symbols.
- B) Determining the meaning of individual words.
- C) Parsing sentences to identify their structure.
- D) Assigning words to predefined categories.

Answer: A

23. What does Named Entity Recognition (NER) in NLP identify?

- A) The sentiment behind a sentence.
- B) Specific entities like names of people, organizations, and locations.
- C) The grammatical structure of a sentence.
- D) The overall topic of a text.

Answer: B

24. In the context of NLU, what is the challenge of "word sense disambiguation"?

- A) Determining the grammatical structure of sentences.
- B) Identifying the correct meaning of a word based on its context.
- C) Parsing large volumes of data.
- D) Translating words from one language to another.

Answer: B

25. Which of the following is an example of an NLU difficulty in the sentence: "He saw her duck"?

- A) Determining the tense of the sentence.
- B) Understanding the meaning of the word "duck" based on context.
- C) Identifying the named entities.
- D) Analyzing the sentence's syntax.

Answer: B

26. In NLP, what is "part-of-speech tagging"?

- A) Assigning each word in a sentence its corresponding part of speech (e.g., noun, verb).
- B) Splitting a sentence into words or tokens.
- C) Analyzing the sentiment of the text.
- D) Identifying named entities in the text.

Answer: A

27. What is "sentiment analysis" in NLP?

- A) Determining the grammatical structure of sentences.
- B) Identifying and classifying opinions and emotions expressed in a text.
- C) Identifying named entities like people, places, or organizations.
- D) Breaking a text into individual words or tokens.

Answer: B

28. In the context of NLP, what is 'Chunking'?

- A) The process of converting a sentence into smaller meaningful groups, such as noun phrases.
- B) Splitting words into individual characters.
- C) Identifying the main subject and predicate in a sentence.
- D) Classifying text into different topics.

Answer: A

29. What is the purpose of the 'Named Entity Recognition' (NER) step in NLP?

- A) To break down the text into smaller pieces.
- B) To identify and classify entities like names of people, organizations, and locations.
- C) To identify the emotional tone of a text.
- D) To detect grammatical errors in the text.

Answer: B

30. How do you install the NLTK library in Python?

- A) pip install nltk
- B) install nltk
- C) pip install nltk.data
- D) install nltk package

Answer: A

31. Which of the following commands is used to download NLTK's data once NLTK is installed?

- A) nltk.download()
- B) download nltk.data
- C) nltk.get()
- D) nltk.data.fetch()

Answer: A

32. What is the purpose of the nltk.corpus package in NLTK?

- A) To install external libraries.
- B) To provide access to various textual corpora for NLP tasks.
- C) To handle machine learning tasks.
- D) To create graphical models.

Answer: B

33. Which NLTK function is used for word tokenization?

- A) nltk.tokenize.word_tokenize()
- B) nltk.tokenize()
- C) nltk.word_tokenizer()
- D) nltk.tokenize.split()

Answer: A

34. Which of the following is an example of Lemmatization in NLTK?

- A) nltk.stem.PorterStemmer()
- B) nltk.stem.WordNetLemmatizer()
- C) nltk.tokenize.word_tokenize()
- D) nltk.parse.syntax()

Answer: B

35. Which of the following NLTK modules is used for chunking text into meaningful groups?

- A) nltk.chunk
- B) nltk.tokenize
- C) nltk.parse
- D) nltk.stem

Answer: A

36. What is the purpose of "Named Entity Chunking"?

- A) To group words into parts of speech.
- B) To identify named entities such as people, locations, and organizations in the text.
- C) To break down text into sentences.
- D) To remove unimportant words.

Answer: B

37. Which of the following is a limitation of the Bag of Words (BoW) model?

- A) It considers the order of words.
- B) It does not consider the semantic meaning of words.
- C) It only works with short texts.
- D) It requires complex computational resources.

Answer: B

38. Which of the following best defines Time Series data?

- A) A set of data points ordered by time.
- B) A random set of data points.
- C) Data points ordered by categories.
- D) Data points that are unrelated.

Answer: A

39. Why is analyzing time series data important?

- A) It helps understand changes over time.
- B) It ignores historical data to predict future outcomes.
- C) It is only useful for financial data.
- D) It does not require any preprocessing.

Answer: A

40. Which of the following is the command to install the Pandas library in Python?

- A) pip install pandas
- B) pip install numpy
- C) pip install matplotlib
- D) pip install scipy

Answer: A

41. In a time series dataset, what does .resample() in Pandas do?

- A) Resamples data by a given frequency.
- B) Calculates cumulative statistics.
- C) Filters the time series data.
- D) Splits data into training and testing sets.

Answer: A

42. What does .rolling() function do in Pandas?

- A) It creates a moving window view of the data.
- B) It sorts the data in ascending order.
- C) It splits the data by time period.
- D) It finds the trend in time series data.

Answer: A

43. In time series analysis, what does autocorrelation measure?

- A) The relationship between past and future values in a time series.
- B) The correlation between two different time series.
- C) The sum of all values in a series.
- D) The variance of the data over time.

Answer: A

44. Which of the following Python libraries is commonly used to work with Hidden Markov Models (HMMs)?

- A) hmmlearn
- B) sklearn
- C) matplotlib
- D) keras

Answer: A

45. In the context of HMM, what does the term 'hidden state' refer to?

- A) A part of the time series data that can be directly observed.
- B) A state that is inferred and not directly observed from the data.
- C) The final output of the model.
- D) A variable that is not relevant for analysis.

Answer: B

46. Which type of chart is commonly used to visualize stock market data?

- A) Line chart
- B) Scatter plot
- C) Pie chart
- D) Histogram

Answer: A

47. What is a common statistical technique used to analyze stock price volatility?

- A) Moving averages
- B) Bollinger Bands
- C) Linear regression
- D) Fourier Transform

Answer: B

48. Which of the following is NOT typically used when analyzing stock market data with Python?

- A) Pandas for data manipulation
- B) NumPy for numerical calculations
- C) Matplotlib for visualization
- D) OpenCV for image processing

Answer: D

49. In building a speech recognizer, which of the following is a crucial step for accurate transcription?

- A) Preprocessing the audio signal to remove noise.
- B) Visualizing the audio waveform.
- C) Running deep learning models on the raw audio data.
- D) Using high-level models without preprocessing.

Answer: A

50. What type of plot is often used to visualize an audio waveform?

- A) Histogram
- B) Line plot
- C) Scatter plot
- D) Spectrogram

Answer: B

51. Which of the following is the correct method to visualize an audio file using a spectrogram?

- A) `plt.plot(audio_signal)`
- B) `librosa.display.specshow()`
- C) `plt.hist(audio_signal)`
- D) `plt.imshow(audio_signal)`

Answer: B

52. Which of the following is the typical output when an audio signal is transformed into the frequency domain?

- A) Spectrogram
- B) Waveform
- C) Time series data
- D) Histogram

Answer: A

53. In signal processing, how is a monotone signal often represented?

- A) As a time series with a constant amplitude.
- B) As a waveform with fluctuating frequencies.
- C) As a frequency domain plot with multiple peaks.
- D) As a time series with variable pitch.

Answer: A

54. Which feature extraction technique is based on mimicking the human ear's perception of sound?

- A) Mel-frequency cepstral coefficients (MFCC)
- B) Fourier Transform
- C) Linear Predictive Coding (LPC)
- D) Z-score normalization

Answer: A

55. Which function in librosa is used to extract MFCCs from an audio signal?

- A) librosa.feature.mfcc()
- B) librosa.feature.extract()
- C) librosa.feature.spectrogram()
- D) librosa.mfcc()

Answer: A

56. What is the role of a speech recognition model during real-time processing?

- A) It transcribes spoken words into text.
- B) It visualizes the audio signal.
- C) It compresses the audio data for faster transmission.
- D) It detects emotions from the speech.

Answer: A

57. Which of the following is NOT a challenge in spoken word recognition?

- A) Variability in speech patterns across different speakers.
- B) Background noise and distortions.
- C) Speech rate and accent differences.
- D) The ability to extract the frequency spectrum.

Answer: D

58. In the A algorithm, which two values are combined to determine the next state to explore?*

- A) Heuristic value and the cost to reach the current state.
- B) Heuristic value and random value.
- C) Cost to reach the goal and the number of steps taken.
- D) Heuristic value and the time complexity.

Answer: A

59. Which of the following search algorithms is based on heuristic search?

- A) Breadth-first search
- B) Depth-first search
- C) A* search
- D) Dijkstra's algorithm

Answer: C

60. What distinguishes informed search algorithms from uninformed search algorithms?

- A) Informed search algorithms do not use any domain-specific information.
- B) Informed search algorithms use a heuristic to make decisions about which path to follow.
- C) Informed search algorithms are slower than uninformed algorithms.
- D) Informed search algorithms are less efficient.

Answer: B

61. Which of the following search strategies is considered an informed search?

- A) Depth-First Search
- B) Breadth-First Search
- C) A* Search
- D) Uniform Cost Search

Answer: C

62. Which of the following is an example of a real-world problem that can be solved using constraint satisfaction techniques?

- A) Crossword puzzles
- B) Chess game playing
- C) Tic-tac-toe
- D) Sorting numbers

Answer: A

63. Which of the following problems can be solved using a constraint satisfaction problem (CSP) framework?

- A) Scheduling classes in a university
- B) Finding the most efficient sorting algorithm
- C) Searching for the shortest path in a graph
- D) Implementing a machine learning algorithm

Answer: A

64. Which of the following is a common technique used to solve constraint satisfaction problems?

- A) Dynamic programming
- B) Backtracking search
- C) Genetic algorithms
- D) Greedy search

Answer: B

65. Which of the following is a common feature of search algorithms in AI?

- A) They use the concept of nodes and edges to represent the problem space.
- B) They only explore a small portion of the search space.
- C) They require no computational resources.
- D) They are primarily used for sorting data.

Answer: A

66. Which of the following search strategies is a combinatorial search method?

- A) Minimax Algorithm
- B) Breadth-First Search
- C) Depth-First Search
- D) A* Search

Answer: B

67. The Minimax algorithm is primarily used in:

- A) Games with a zero-sum outcome.
- B) Searching unstructured data.
- C) Optimization of non-game related problems.
- D) Linear regression models.

Answer: A

68. In the Minimax algorithm, the goal is to:

- A) Maximize the player's own score and minimize the opponent's score.
- B) Minimize the player's own score and maximize the opponent's score.
- C) Make random decisions at each level.
- D) Minimize the number of moves.

Answer: A

69. In Alpha-Beta pruning, what does the "alpha" value represent?

- A) The maximum value the maximizer can guarantee so far.
- B) The minimum value the minimizer can guarantee so far.
- C) The average value of all possible moves.
- D) The best possible outcome for the opponent.

Answer: A

70. Which of the following is a benefit of Alpha-Beta Pruning over standard Minimax?

- A) It ensures that the game is played optimally.
- B) It speeds up the decision-making process by eliminating branches that do not need to be explored.
- C) It simplifies the decision tree.
- D) It ensures that the worst possible moves are avoided.

Answer: B

71. Which of the following is an advantage of the Negamax algorithm over Minimax?

- A) It is simpler to implement by using the same evaluation function for both players.
- B) It is faster than Alpha-Beta pruning.
- C) It guarantees finding the optimal solution in all cases.
- D) It uses a probabilistic approach to decision making.

Answer: A

A Bot to Play Last Coin Standing

72. In the "Last Coin Standing" game, the goal for the bot is to:

- A) Maximize the number of coins it collects.
- B) Ensure that it always picks the last coin.
- C) Minimize the number of coins it takes during each turn.
- D) Predict the moves of the opponent without any constraints.

Answer: B

73. Which strategy can a bot use to win at "Last Coin Standing"?

- A) Always leave an even number of coins for the opponent.
- B) Randomly pick coins without any strategy.
- C) Always leave an odd number of coins for the opponent.
- D) Ignore the number of coins left and make moves based on heuristics.

Answer: A

74. What is a reward function in reinforcement learning?

- A) A function that calculates the actions to be taken by the agent.
- B) A function that gives the agent feedback based on its actions, helping it learn.
- C) A function that adjusts the state of the environment.
- D) A function used to evaluate the performance of the agent's environment.

Answer: B

75. Which of the following is NOT a key component of a reinforcement learning setup?

- A) Agent
- B) Environment
- C) Data Labeling
- D) Reward Function

Answer: C

76. What is the term for the sequence of states and actions taken by an agent in reinforcement learning?

- A) Policy
- B) Trajectory
- C) Model
- D) Dataset

Answer: B

77. What is the purpose of the "state" in reinforcement learning?

- A) To define the available actions for the agent.
- B) To represent the current situation of the environment that the agent perceives.
- C) To store the rewards received by the agent.
- D) To calculate the agent's next action.

Answer: B

Constructing an Environment with Python

78. Which Python library is commonly used for building reinforcement learning environments?

- A) TensorFlow
- B) OpenAI Gym
- C) NumPy

D) Keras

Answer: B

79. In OpenAI Gym, what function is used to create a new environment?

- A) gym.make()
- B) gym.initialize()
- C) gym.create()
- D) gym.reset()

Answer: A

80. Which of the following represents the typical structure of an environment in reinforcement learning?

- A) The agent interacts with a fixed, predefined environment without any feedback.
- B) The environment generates random actions without any agent involvement.
- C) The environment reacts to the agent's actions and provides a new state and reward.
- D) The environment is a static set of states with no interaction with the agent.

Answer: C

81. In Python, what type of object is an environment in reinforcement learning (when using OpenAI Gym)?

- A) A list of predefined actions.
- B) An instance of the gym.Env class.
- C) A simple function that returns rewards.
- D) A class that generates the agent's learning model.

Answer: B

82. What does the Q-value (or action-value) represent in Q-learning?

- A) The state of the agent.
- B) The reward that can be obtained by taking a particular action from a particular state.
- C) The environment's response to the agent's actions.
- D) The probability of the agent's actions succeeding.

Answer :B

83. What is the purpose of the "policy" in reinforcement learning?

- A) It defines the actions the agent will take at each state to maximize rewards.
- B) It randomly selects actions for the agent to take.
- C) It determines how the environment reacts to the agent's actions.
- D) It evaluates the learning agent's performance.

Answer: A

What are Genetic Algorithms?

84. What is the primary inspiration behind Genetic Algorithms (GAs)?

- A) Quantum computing principles
- B) Natural selection and evolution in biology
- C) Gradient descent in machine learning
- D) Random walk theory

Answer: B

85. Which of the following is a key component of a Genetic Algorithm?

- A) Population, fitness function, and crossover
- B) Learning rate and activation function
- C) Heuristic estimation and depth-first search
- D) Backpropagation and loss minimization

Answer: A

86. What does the fitness function evaluate in a Genetic Algorithm?

- A) The speed of the algorithm's execution
- B) How well a solution solves the given problem
- C) The memory usage of the algorithm
- D) The number of iterations required

Answer: B

87. In the context of optimization, what role does the crossover operation play in GAs?

- A) It introduces random noise to prevent convergence
- B) It combines features of two parent solutions to create offspring
- C) It eliminates the weakest individuals from the population
- D) It evaluates the fitness of the current population

Answer: B

88. How does mutation contribute to solving optimization problems in GAs?

- A) It ensures the population remains identical over generations
- B) It introduces diversity to avoid getting stuck in local optima
- C) It calculates the fitness of each individual
- D) It selects the top-performing solutions

Answer: B

89. Which of the following code snippets reads an image using OpenCV?

- A) `cv2.read('image.jpg')`
- B) `cv2.imread('image.jpg')`
- C) `cv2.show('image.jpg')`
- D) `cv2.load('image.jpg')`

Answer: B

90. Which function from OpenCV is used to display an image in a window?

- A) `cv2.show()`
- B) `cv2.display()`
- C) `cv2.imshow()`
- D) `cv2.plot()`

Answer: C

91. Which function is used to write an image to a file using OpenCV?

- A) `cv2.save()`
- B) `cv2.write()`
- C) `cv2.imwrite()`
- D) `cv2.store()`

Answer: C

92. Which of the following techniques is often used to improve the quality of an image for further analysis?

- A) Histogram equalization
- B) Edge detection
- C) Image segmentation
- D) Optical character recognition

Answer: A

93. What does "thresholding" do in image processing?

- A) Converts an image into grayscale.
- B) Detects the boundaries of an image.
- C) Converts an image into a binary format based on intensity levels.
- D) Changes the contrast of the image.

Answer: C

94. What is the advantage of using the HSV color space over the RGB color space for image processing tasks?

- A) HSV is less sensitive to changes in lighting conditions.
- B) HSV has a simpler representation of color.
- C) RGB is easier to compute and process.
- D) RGB provides more distinct colors than HSV.

Answer: A

95. Which of the following is an activation function commonly used in Artificial Neural Networks?

- A) Linear function
- B) Sigmoid function
- C) Polynomial function
- D) Logarithmic function

Answer: B

96. In an Artificial Neural Network, what is the role of the 'weights'?

- A) To adjust the output of the neuron.
- B) To prevent overfitting.
- C) To control the learning rate.
- D) To update the activation function.

Answer: A

97. Which OpenCV function is used to perform edge detection using the Canny method?

- A) `cv2.Canny()`
- B) `cv2.edges()`
- C) `cv2.detectEdges()`
- D) `cv2.findContours()`

Answer: A

98. In edge detection, which of the following methods is often applied to reduce noise before detecting edges?

- A) Sobel filtering
- B) Gaussian blur
- C) Thresholding

D) Resizing

Answer: B

99. In a neural network used for linear regression, what is typically the activation function of the output layer?

A) ReLU

B) Sigmoid

C) Tanh

D) Linear

Answer: D

100. In an ANN for linear regression, what is the purpose of the loss function?

A) To adjust the weights of the network during training.

B) To predict the output values.

C) To compute the gradient for backpropagation.

D) To introduce regularization.

Answer: A